Sentence-Level Factuality Classification in Brazilian Portuguese News

An NLP Pipeline using Traditional Machine Learning

FACTNEWS DATASET | BRAZILIAN PORTUGUESE | 3-CLASS CLASSIFICATION

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Dataset: FactNews (Vargas et al., 2023)

Source

GitHub · Francielle Vargas · 300 Brazilian Portuguese news articles

Scale

6,191 sentences · 7 columns · sentence-level annotation

Quality

No missing values, no duplicates · human-annotated·

Label 0

Factual

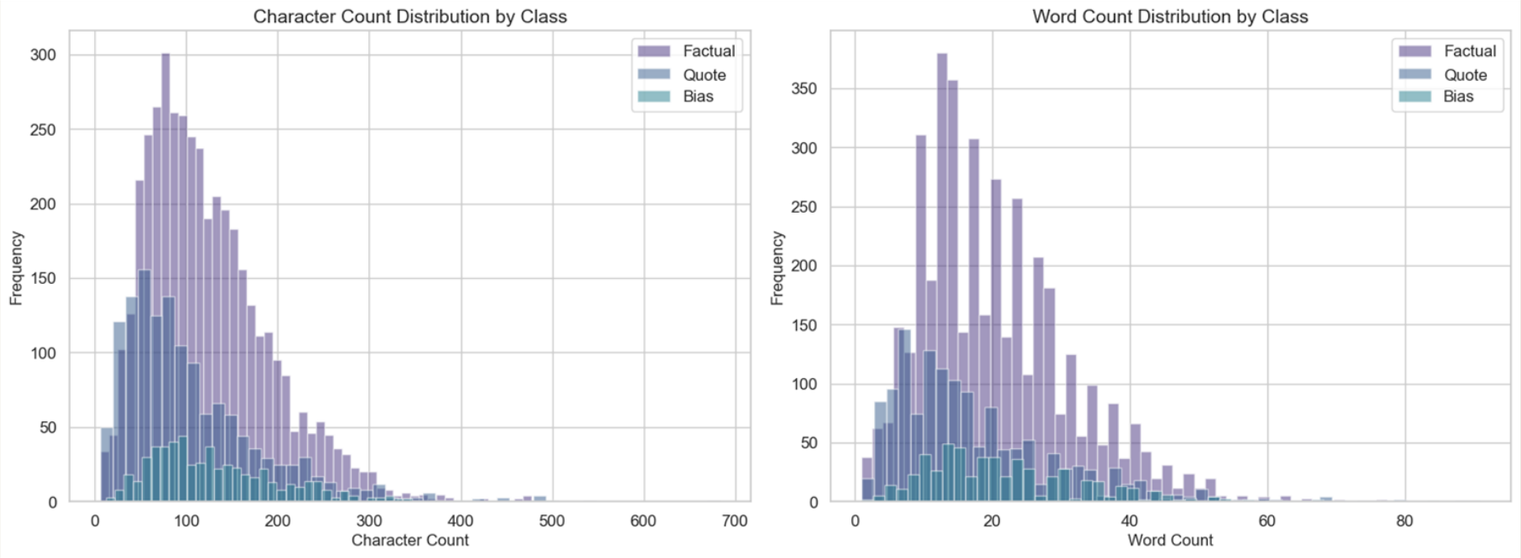
Label –1

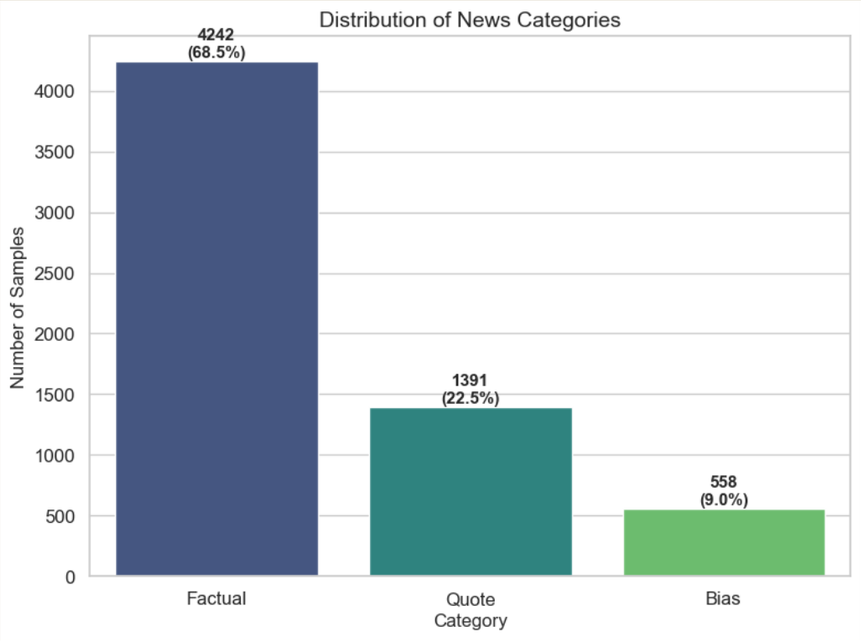
Quote

Label 1

Bias

EDA: Class Imbalance & Length Patterns





Severe Imbalance

Bias is 7.6× smaller than Factual → accuracy is not a valid metric

Primary Metric

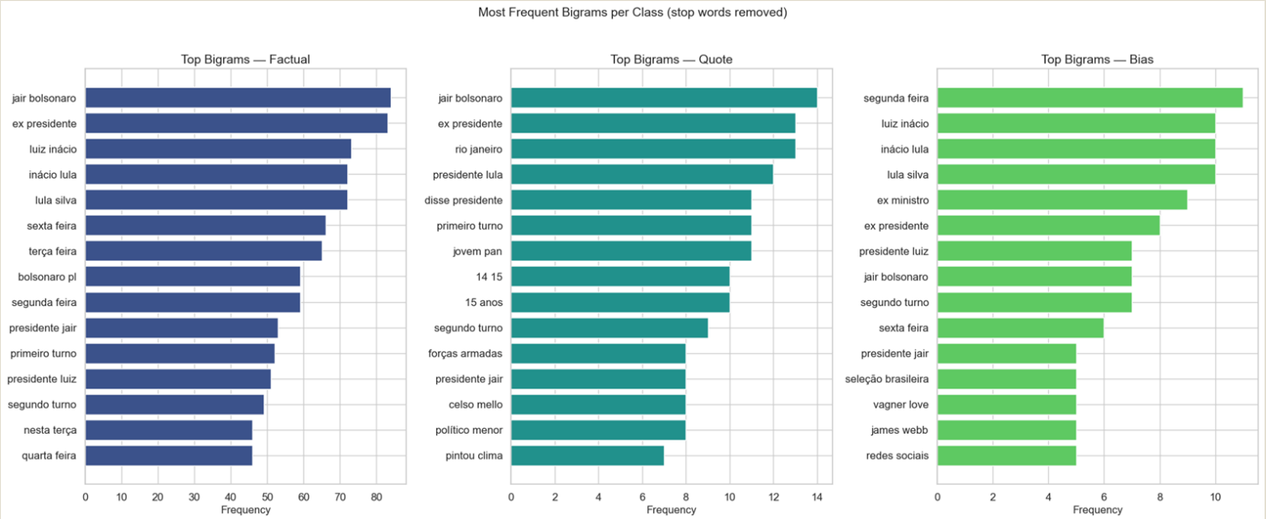
Macro-F1 — treats all three classes equally regardless of frequency

Length Patterns

Bias sentences longest on average · Factual has widest spread

Splitting Strategy

Stratified 80/20 train/test split — mandatory given imbalance



EDA: Vocabulary Analysis & N-gram Patterns

Quote Class

Speech-act phrases · attribution verbs: *disse, afirmou* · indirect speech markers

Raw Vocabulary

~44,000+ unique tokens · dominated by Portuguese stop words (*de, a, o, que*)

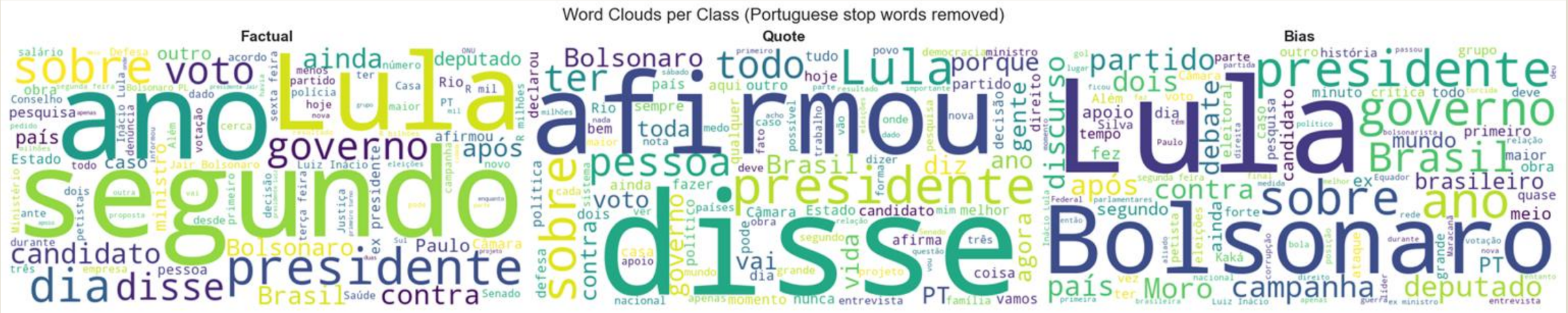
Factual Class

Entity names · descriptive noun phrases · neutral register

Bias Class

Evaluative and subjective constructions · sentiment-laden vocabulary

- Stop words dominate raw text → removal is critical · TTR measured per class to quantify lexical diversity · ~68% vocabulary reduction after preprocessing



Preprocessing: Portuguese-Aware NLP Pipeline

Lowercasing
Reduce vocabulary size

Example: [Factual]
BEFORE: A margem de erro é dois pontos para mais ou para menos.

Remove non-alphabetic
Keep letters and diacritics

Tokenisation
Split text into words

Stop-word removal
NLTK Portuguese list (210)

RSLP Stemming
Portuguese-specific stems

Example: [Factual]
AFTER: marg err doi pont menos

Why RSLP?
Handles Portuguese morphology correctly · Porter stemmer is English-only and corrupts PT tokens

Vocabulary Reduction
~44K unique tokens (raw) → ~14K unique tokens (cleaned)

Feature Representations: Sparse vs. Dense

Method	Dimensions	Sparsity
BoW (CountVectorizer)	5,000	~99%
TF-IDF (1-2 gram)	5,000	~99%
FastText	300	0%

Undersampling on training set: majority classes downsampled to ~446 samples each but test set retains original distribution for real-world evaluation

Bag of Words

- Raw term frequencies
- effective for lexical “trigger words”

TF-IDF

- Weights by importance
- captures bigrams better
- Sublinear term frequency (TF) scaling

FastText

- Pre-trained CC + Wikipedia PT
- 300-dimensional dense vectors
- handles out-of-vocabulary terms (OOV) and portuguese verb conjugations via subword information

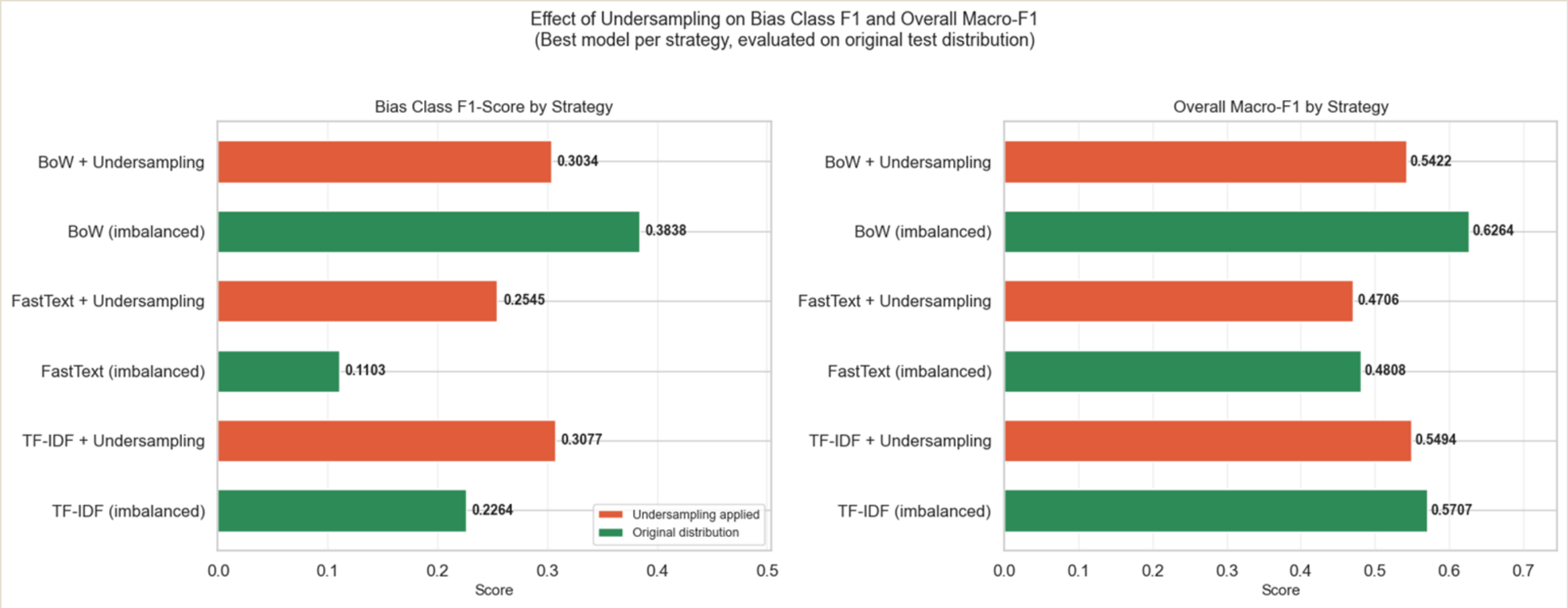
Experimental Results: Model Comparison

Key Insight

Specific lexical trigger words are more discriminative than semantic embeddings for factuality

Best Model

Naïve Bayes + BoW · Macro-F1 = **0.6264**



Results: Sparse vs. Dense Imbalance Strategies

Accuracy-F1 Gap

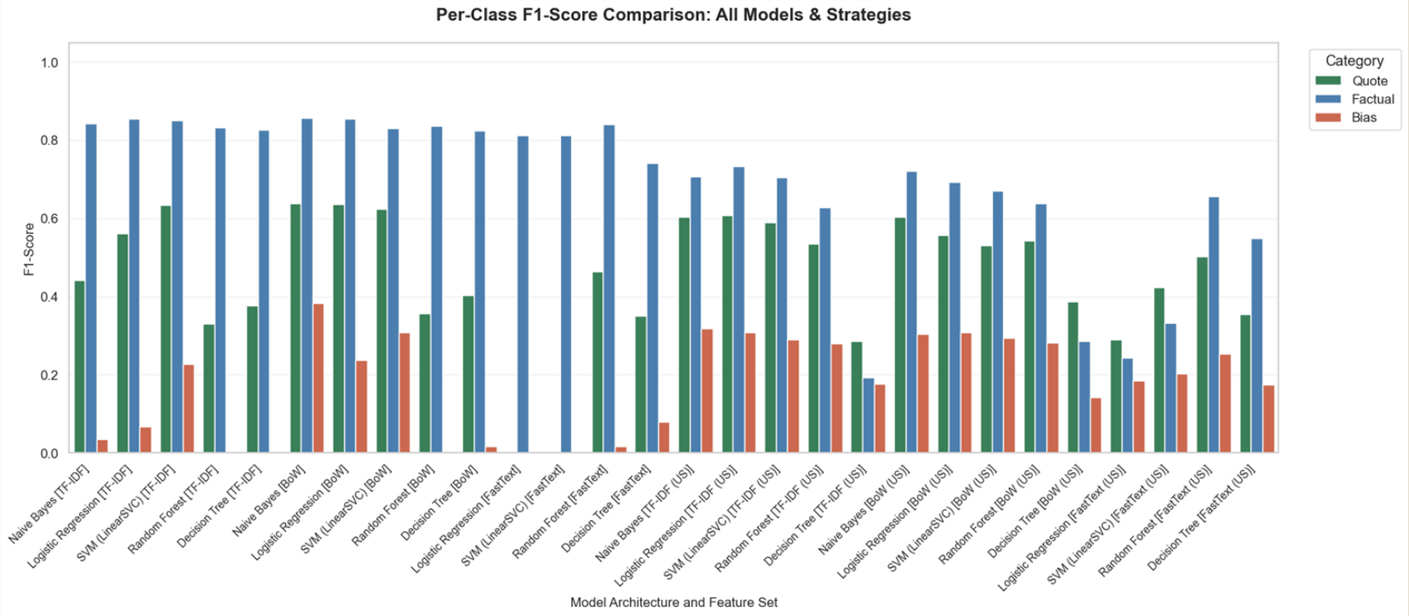
~77% accuracy vs. ~62% Macro-F1 - models over-predict the majority Factual class

Undersampling Effect

Improves Bias-class recall · small cost to overall performance.

Dense Underperforms

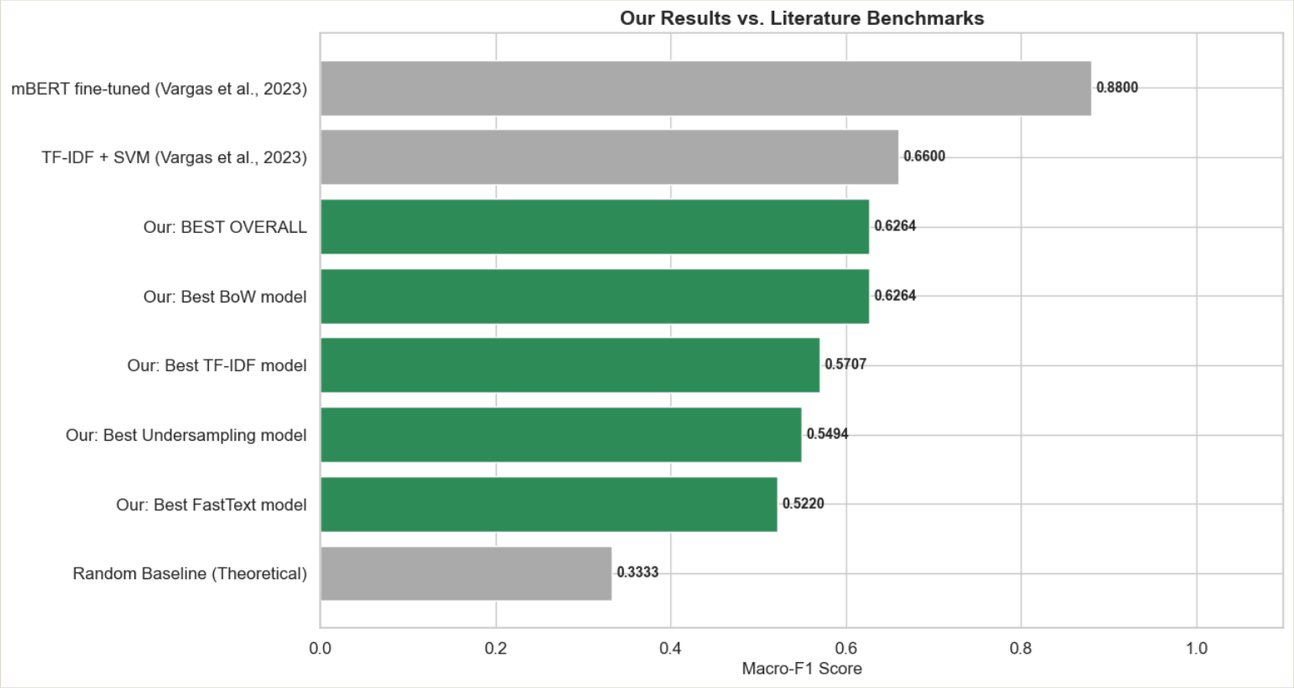
Averaging FastText vectors blurs structural and lexical markers needed for factuality detection



vs. Vargas et al. (2023)

mBERT fine-tuned: 88% F1 · Traditional ML reported: 70–80% · Our result (0.6264) is competitive

Error Analysis: Where Models Fail



Bias ↔ Factual

Primary error · subtle bias closely resembles factual reporting even for human annotators

Quote ↔ Factual

Indirect speech without explicit attribution markers · quotation marks removed during cleaning

t-SNE Confirms

Quote forms a distinct cluster · Factual/Bias overlap significantly in semantic space

Bottleneck

Bias class remains the primary performance bottleneck across all models

Results comparison with the paper:
https://doi.org/10.26615/978-954-452-092-2_127

Conclusion

01

Best Result

Naïve Bayes + BoW · Macro-F1 = **0.6264** · lexical anchors dominate over semantics

02

PT Preprocessing is Essential

RSLP stemmer + Portuguese stop words · ~68% vocabulary reduction

03

Sparse > Dense

BoW/TF-IDF outperform averaged FastText for factuality detection

04

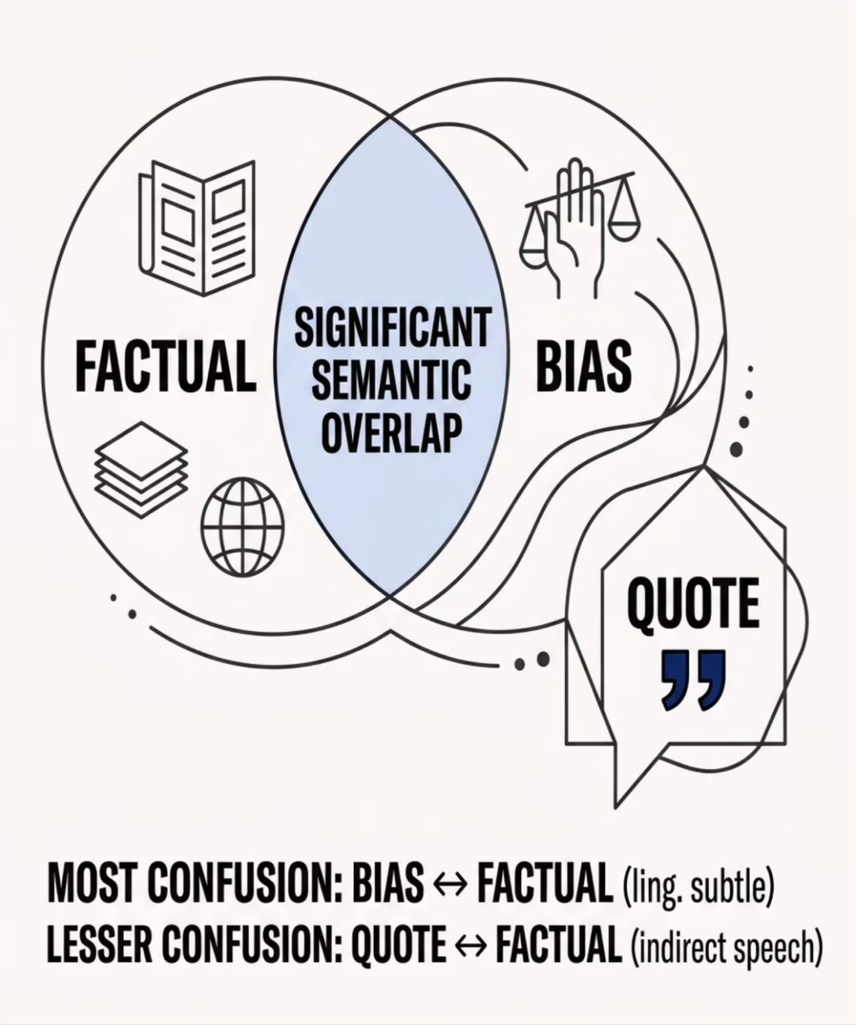
Imbalance is the Core Challenge

7.6:1 ratio · Macro-F1 is the most correct metric

Limitations

No deep learning (by design) · averaged FastText loses sentence structure · undersampling reduces training data

Future Work

Sentence-BERT embeddings · ensemble methods · lexicon-based features for Bias detection · Stemming instead of lemmatization

Dataset: FactNews — Vargas et al. (2023) · github.com/franciellevargas/FactNews